The divergence theorem in Lorentzian geometry

The purpose of this note is to provide a short and self-contained derivation of the divergence theorem in Lorentzian geometry, with particular attention to sign conventions and boundary.

1 Stokes' theorem

Recall that, on an oriented manifold $\mathcal{M}^n$, the integral of an n-form ω is defined as follows. One first fixes an orientation by choosing a nowhere-vanishing n-form ϵ. Then ω can be written as $\omega = f\,\epsilon$ for some function f. The integral of ω is then defined by integrating f in local coordinates, using a partition of unity if necessary.

If $\mathcal{M}$ is a manifold with boundary, then its boundary ∂M carries a canonical induced orientation. At each point, an ordered basis of $T_p(\partial M)$ is declared to be positively oriented if, after pre-appending an outward-pointing transversal vector, it becomes a positively oriented basis of T_pM.

Stokes' theorem allows one to pass from integrals over $\mathcal{M}$ to integrals over ∂M. More precisely, for any $(n-1)$-form ω, one has

$$\int_M d\omega = \int_{\partial M} \omega,$$

where the orientation on ∂M is the one induced from $\mathcal{M}$, and ω on the boundary is understood as the pullback via the inclusion map.

It is important to emphasize that none of these constructions depend on the presence of a metric. The metric will only be used later to define a volume form, the divergence of a vector field, and to rewrite the terms appearing in Stokes' theorem in a more convenient form.

2 The "normal-free" approach and Cartan's homotopy formula

Recall that the Lie derivative is a derivation defined independently of any metric structure. The flow generated by a vector field X (the "direction of derivation") provides a way to compare tensors at different points, allowing one to define the derivative via an appropriate limit of difference quotients.

While the Lie derivative can be defined on arbitrary tensor fields, its action on differential forms enjoys a particularly useful relationship with the exterior derivative d and the contraction operator ι_X.

Lemma 1 (Cartan's homotopy formula)**.** *Let ω be a differential form and X a vector field on $\mathcal{M}$. Then*

$$\mathcal{L}_X\,\omega = d\iota_X\omega + \iota_X d\omega,$$

where ι_X, the interior product, maps a $(k+1)$-form ω to the k-form defined by

$$(\iota_X\omega)(Y_1,\dots,Y_k)=\omega(X,Y_1,\dots,Y_k).$$

In particular, if ω is a top-degree form, then $d\omega=0$, and we are left with the relation $\mathcal{L}_X\,\omega=d\iota_X\omega$.

Applying Stokes' theorem to $\iota_X\omega$ on an n-dimensional manifold $\mathcal{M}$, we obtain

$$\int_R \mathcal{L}_X\,\omega=\int_{\partial R}\iota_X\omega.$$

This is the version of Stokes' theorem that we will use to derive the divergence theorem.

3 The divergence theorem

We now introduce the metric structure on M. Recall that a metric g, together with a choice of orientation, determines a volume form ϵ. In positively oriented coordinates $x^1,\dots,x^n$, one has

$$\epsilon=\sqrt{|g_{ij}|}\,dx^1\wedge\cdots\wedge dx^n.$$

Intuitively, the Lie derivative $\mathcal{L}_X\,\epsilon$ measures the infinitesimal change of volume of a region transported along the flow of X. It is therefore natural that it is related to the divergence of X.

Lemma 2. *Let X be a vector field and denote by $\nabla^\alpha X_\alpha$ its divergence. Then*

$$\mathcal{L}_X\,\epsilon=(\nabla^\alpha X_\alpha)\,\epsilon.$$

Proof. This follows from Cartan's homotopy formula. Let $\mu=\sqrt{|g_{ij}|}$. Then

$$\begin{aligned}
\mathcal{L}_X\,\epsilon&=d(\iota_X\epsilon)\\
&=d\left(\sum_{\alpha=1}^{n}(-1)^{\alpha+1}X^\alpha\mu\,dx^1\wedge\cdots\widehat{dx^\alpha}\cdots\wedge dx^n\right)\\
&=\sum_{\alpha=1}^{n}\partial_\alpha(\mu X^\alpha)\,dx^1\wedge\cdots\wedge dx^n\\
&=\frac{1}{\mu}\partial_\alpha(\mu X^\alpha)\,\epsilon,
\end{aligned}$$

where $\widehat{dx^\alpha}$ denotes omission of the factor dx^α. The quantity $\frac{1}{\mu}\partial_\alpha(\mu X^\alpha)$ is the standard expression for the divergence of X. ∎

The divergence theorem therefore takes the form

$$\int_{\mathcal{M}}\operatorname{div}(X)\,\epsilon=\int_{\partial\,\mathcal{M}}\iota_X\epsilon.$$

It remains to make the boundary term more explicit. Let $p\in\partial\,\mathcal{M}$, and choose a basis $e_2,\dots,e_n$ of $T_p\partial\,\mathcal{M}$. Let e_1 be a vector completing it to a basis of T_pM. Writing $X=X^ie_i$, one has

$$\iota_X\epsilon(e_2,\dots,e_n)=X^1\,\epsilon(e_1,\dots,e_n).$$

If $T_p\partial R$ is spacelike or timelike, a natural choice is to take e_1 to be a normal vector. In this case,

$$X^1 = \pm g(X, e_1),$$

where the sign depends on the causal character of e_1, and $\iota_{e_1}\epsilon$ coincides (up to orientation) with the induced volume form on ∂R.

If instead $T_p\partial R$ is null, the normal direction lies in the tangent space and cannot be used to complete the basis. In this case, we introduce a null frame $(\underline{L}, L, e_1, \ldots, e_{n-2})$, where:

- L is (up to scale) the null normal/generator of the hypersurface,
- $\underline{L}$ is a null vector transverse to the hypersurface such that $g(L, \underline{L}) = -1$,
- e_i are orthonormal spacelike vectors tangent to ∂R.

Usually, L is chosen to be future-directed with a convenient normalization.

Writing $X = X^L L + X^{\underline{L}}\underline{L} + X^i e_i$, one finds

$$X^{\underline{L}} = -g(X, L),$$

and therefore

$$\iota_X\epsilon = -g(X, L)\,\iota_{\underline{L}}\epsilon.$$

It is important to note that, although the choice of L and $\underline{L}$ affects the induced volume form $\iota_{\underline{L}}\epsilon$, the total boundary integral is independent of this choice: after all, it was already defined intrinsically by $\iota_X\epsilon$!

4 A couple of examples

As a first example, one can take energy estimates for solutions to the wave equation on Minkowski spacetime $(\mathbb{R}^{n+1}, m)$. In this case, the covariant wave equation

$$\Box_m\psi = 0$$

is just the usual wave equation

$$\partial_t^2\psi - \Delta\psi = 0.$$

One can attach the *energy-momentum tensor* $T_{\mu\nu}(\psi) = \partial_\mu\partial_\nu\psi - \frac{1}{2}m_{\mu\nu}\partial^\alpha\psi\partial_\alpha\psi$, and given a vector field X^μ, the associated current $J_\mu^{(X)} = T_{\mu\nu}(\psi)X^\nu$. The main interest in this current lies in the fact that, when X^μ is a timelike vector field, the flux of this current controls the energy of the wave ψ. We will take as vector field $T = \partial_t$ and we will apply the divergence theorem to this current over a "reverse truncated conic boundary": take a point (x, t) (the vertex of our cone), a time $r < t$ and consider the domain $\mathcal{C} = \{(y, s) : 0 \le s \le r \text{ and } |y - x| \le |s - t|\}$. $\partial\mathcal{C}$ is comprised of three parts: the two spacelike slices $\Sigma_0 = \{s = 0\} \cap \mathcal{C}$ and $\Sigma_r = \{s = r\} \cap \mathcal{C}$ and the conic boundary $\mathcal{H} = \{|y - x| = |s - t|\} \cap \mathcal{C}$. One can verify that $\nabla^\mu J_\mu = 0$, and so the divergence theorem yields (for $L = T - \nu$ and $\underline{L} = T - \nu$, ν being the normal of the circular sections of the cone)

$$0 = \int_{\Sigma_t}\iota_J\epsilon + \int_{\Sigma_0}\iota_J\epsilon + \int_{\mathcal{H}}\iota_J\epsilon = -\int_{\Sigma_t}J_\mu T^\mu + \int_{\Sigma_0}J_\mu T^\mu - \int_{\mathcal{H}}J^\mu L_\mu\iota_{\underline{L}}\epsilon,$$

and one can see that $\iota_{\underline{L}}\epsilon$ is exactly the volume form (with the right orientation) on the boundary manifold – just rotate $\underline{L}$ to T. So we have derived the energy balance:

$$\int_{\Sigma_t} J_\mu T^\mu + \int_{\mathcal{H}} J^\mu L_\mu = \int_{\Sigma_0} J_\mu T^\mu;$$

from the positivity property of J – namely, J is positive when applied to a causal vector – the second term can be dropped to obtain the *domain of dependency property*:

$$\int_{\Sigma_t} J_\mu T^\mu \leq \int_{\Sigma_0} J_\mu T^\mu.$$